Q1. Data Input Output

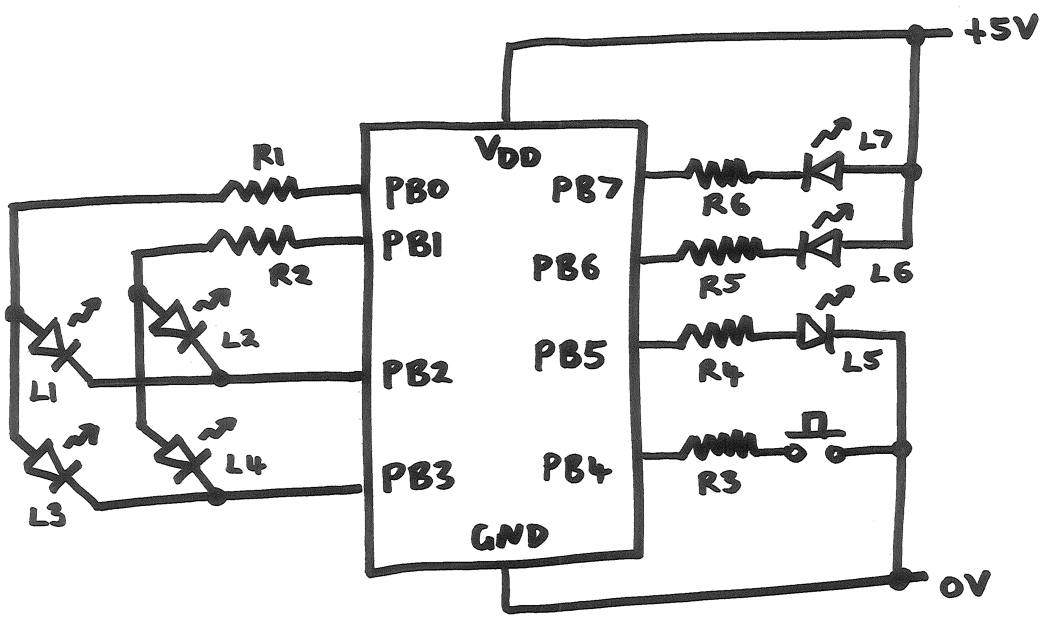


Figure 1: Schematic showing seven LEDs and a pushbutton connected to an ATmega32u2 microcontroller.

An ATmega32u2 microcontroller has seven LEDs (L1 to L7) and a pushbutton connected as shown in Figure 1. Consider this schematic and answer the following questions assuming that the CPU is clocked at 8 MHz. Hint, an LED lights if current flows from the anode (indicated by the triangle) to the cathode.

(a) What **hexadecimal** value needs to be stored in the `DDRB` register?

(b) Which, if any, of the LEDs L5–L7 are active low?

(c) What is the purpose of R3?

(d) Suggest why the pushbutton might not operate? Assume it is not broken!

(e) Write a single C statement that will turn on the LED L7 and not affect the states of the other LEDs.

(f) What **decimal** value needs to be stored in the `PORTB` register so that only the L1 and L3 LEDs are on?

Q2. Data Input Output 2

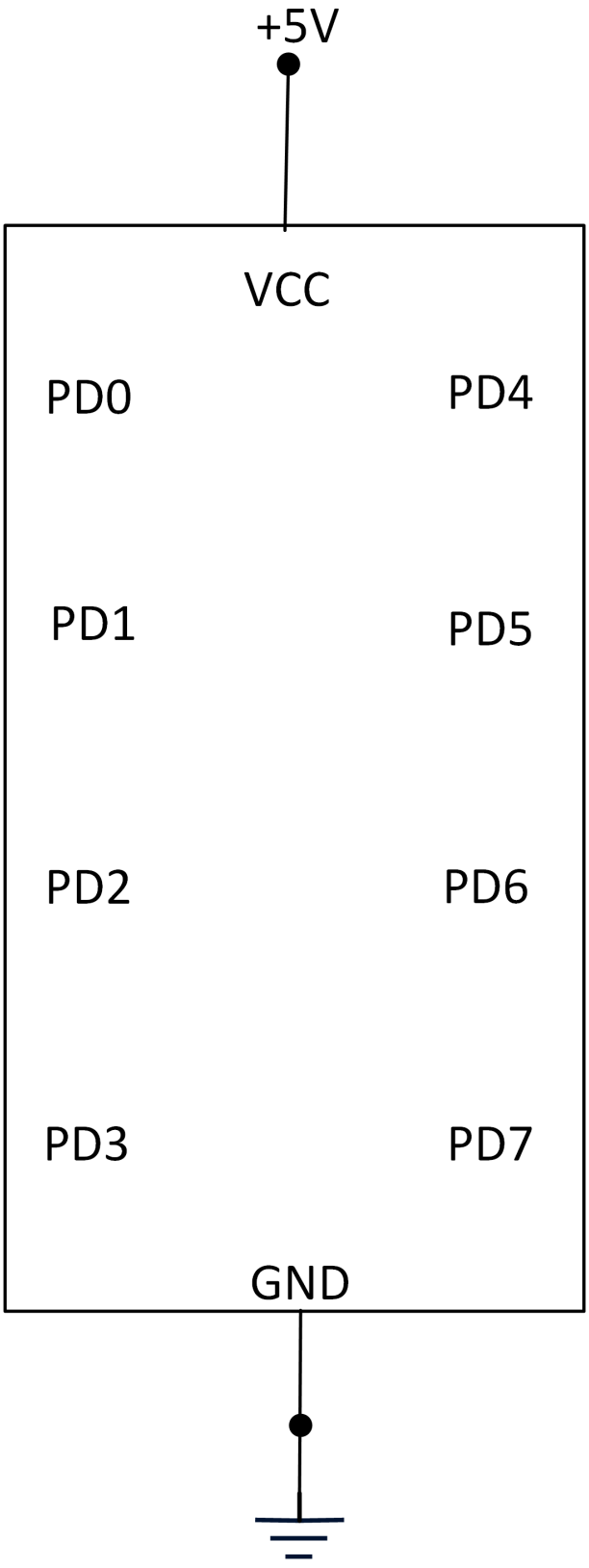


Figure 2: Schematic showing an ATmega32u2 microcontroller to be filled in for Question 2.

Port D of an ATmega32u2 micrcocontroller is connected to one switch and five LEDs (L1, L2, L3, L4, L5), but the input/output pins for these are unknown. Three of the five LEDs are each connected to one PIO pin only. Two of the five LEDs are each connected across two PIO pins. No pins are unused. Answer the following:

(a) The following line evaluates to `True` when the switch is pushed:

```
if ((PIND & (1<<3))==0)
```

Draw the switch and any required electronics for correct operation into Figure 2.

(b) What **decimal** value should be stored in the Data Direction Register for Port D (DDRD)?

(c) L1 is an active low LED and is toggled by:

```
PORTD ^= (1<<0);
```

Draw and label L1 and any required electronics for correct operation into Figure 2.

(d) L2 and L3 are both active high LEDs and are the only LEDs turned on by:

```
PORTD = 0x C1;
```

Draw L2 and L3 and any required electronics for correct operation into Figure 2. (It does not matter which one you label L2 or L3).

(e) LEDs L4 and L5 both connect across two adjacent PIO pins. L4 and L5 are the only LEDs turned on by:

```
PORTD = 35;
```

Draw L4 and L5 and any required electronics for correct operation into Figure 2. (It does not matter which one you label L4 or L5).